Chapter 5. Representations of finite groups: Further results 93
§5.1. Frobenius-Schur indicator 93
§5.2. Algebraic numbers and algebraic integers 95
§5.3. Frobenius divisibility 98
§5.4. Burnside's theorem 100
§5.5. Historical interlude: William Burnside and intellectual harmony in mathematics 102
§5.6. Representations of products 107
§5.7. Virtual representations 107
§5.8. Induced representations 107
§5.9. The Frobenius formula for the character of an induced representation 109
§5.10. Frobenius reciprocity 110
§5.11. Examples 112
§5.12. Representations of S_n 112
§5.13. Proof of the classification theorem for representations of S_n 114
§5.14. Induced representations for S_n 116
§5.15. The Frobenius character formula 118
§5.16. Problems 120
§5.17. The hook length formula 121
§5.18. Schur-Weyl duality for $\mathfrak{gl}(V)$ 122
§5.19. Schur-Weyl duality for $GL(V)$ 124
§5.20. Historical interlude: Hermann Weyl at the intersection of limitation and freedom 125
§5.21. Schur polynomials 131
§5.22. The characters of L_λ 132
§5.23. Algebraic representations of $GL(V)$ 133
§5.24. Problems 135
§5.25. Representations of $GL_2(\mathbb{F}_q)$ 135
§5.26. Artin's theorem 144